

ota-android-agent :core — poll/ protocol/delta coverage raise

Date: 2026-06-23 **Scope:** submodules/ota-android-agent :core JVM module only (kotlin.jvm + JUnit5 + JaCoCo). **Command:** gradle :core:test :core:jacocoTestReport (system Gradle 9.5; no gradlew wrapper present). **Evidence:** jacocoTestReport.after.xml (authoritative JaCoCo XML), test_changes.diff.

BEFORE → AFTER (authoritative JaCoCo XML counters)

Package	LINE before	LINE after	BRANCH before	BRANCH after
poll	77.14% (27/35)	100.00% (35/35)	87.50% (35/40)	100.00% (40/40)
protocol	100.00% (148/148)	100.00% (148/148)	85.71% (12/14)	100.00% (14/14)
delta	100.00% (34/34)	100.00% (34/34)	87.50% (28/32)	100.00% (32/32)
:core all	97.98% (389/397)	100.00% (397/397)	95.88% (256/267)	100.00% (267/267)

Note: the conductor context cited 91.18% LINE from an earlier full-module JaCoCo run. The current :core-only run measures 97.98% LINE baseline; both are reported honestly.

Tests added (8 new, all REAL assertions on actual behaviour)

- **poll/PollStateMachineTest.kt (+4):**
 - `nonTerminalFlags_everyIntermediateStateIsNonTerminal` — exercises Polled/Assigned/Downloading/Downloaded/Applying states + asserts `isTerminal == false`.
 - `poll_fromPolledState_isAlsoValid` — covers the previously-untested Polled branch of the `onPoll` check precondition (all three outcomes).
 - `illegalTransition_onPollFromAssigned_throws` — covers the check failure branch on line 65 (`onPoll` from invalid state).
 - `illegalTransition_onApplyResultFromNonApplying_throws` — covers the check failure branch on line 106 (`onApplyResult` from non-Applying).

- **poll/JitterTest.kt** (+2):
 - `defaultRng_isUsedWhenRngOmitted` — exercises the `nextDelayMillis$default` bridge (2-arg form, default `Random.Default`), deterministic via zero jitter.
 - `defaultRng_withJitter_staysWithinBounds` — default RNG with real jitter, asserts `[base, base+jitterMax)`.
- **delta/DeltaApplyDecisionTest.kt** (+3 incl. accessor test):
 - `fullPayload_whenCurrentShaBlank_butVersionPresent` — isolates the `curSha.isEmpty()` operand (line 100) of the malformed `||` chain.
 - `fullPayload_whenOfferBaseShaBlank_butBaseVersionPresent` — isolates the `baseSha.isEmpty()` operand (line 102).
 - `decisionFields_choiceAndReasonAreReadable` — exercises `DeltaDecision.choice/.reason` accessors directly.
- **protocol/CodecRoundTripTest.kt** (+2):
 - `telemetryAck_absentCounters_defaultToZero` — both `?: 0` elvis fallbacks (lines 137/138) when `accepted+rejected` absent.
 - `telemetryAck_partialCounters_onlyAcceptedPresent` — rejected fallback only.

Test result

112 tests, 0 failures, 0 errors, 0 skipped. BUILD SUCCESSFUL.

Honestly-left / unreachable

None left uncovered at the package LINE/BRANCH level — all three target packages reached 100/100.

Sub-line detail (§11.4.6 honesty): in `PollStateMachine.kt` the `data class Failed(reason)` declaration (line 49) and the `isTerminal` getter (line 52) each retain a few **missed instructions** (synthetic `equals/hashCode/toString/copy` bytecode generated by the Kotlin compiler for the data class, and an unreachable instruction in the getter). JaCoCo's LINE and BRANCH counters report these source lines as COVERED (poll LINE = 35/35, BRANCH = 40/40); the residual missed *instructions* are compiler-generated members not invoked by any test and are not contorted-for to hit.

Bug found

None — no source change required; all four edits are test-only.

Files changed (test-only, ota-android-agent)

```
M core/src/test/kotlin/digital/vasic/helix/ota/core/delta/
DeltaApplyDecisionTest.kt
M core/src/test/kotlin/digital/vasic/helix/ota/core/poll/
JitterTest.kt
M core/src/test/kotlin/digital/vasic/helix/ota/core/poll/
```

```
PollStateMachineTest.kt  
M core/src/test/kotlin/digital/vasic/helix/ota/core/protocol/  
CodecRoundTripTest.kt
```